

# Abdulwahab Felemban

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PhD candidate (expected 2027) specializing in multimodal LLMs, VLMs, and VLAs for autonomous driving, motion prediction, and safety-aware planning.

## RESEARCH INTERESTS

- Multimodal LLMs, VLMs, and VLAs for autonomous driving perception, planning, and generation
- Motion prediction, trajectory generation, world models, and structured reasoning in dynamic scenes
- Safety-aware instruction tuning and evaluation for autonomy systems
- Large-scale autonomous-driving data curation, benchmarking, and closed-loop analysis

## PUBLICATIONS

- **Felemban, A.**, Hroub, N., Ding, J., Fu, Y., Hammoud, H. A. A. K., Bezerra, L. C. D., Khan, F. F., Zhang, W., Elhoseiny, M. *DriveGavel: A Zero-Shot VLM Safety Layer for Autonomous Driving Planners* [Under review, AAAI 2027; advanced to Phase 2]
- **Felemban, A.**, Hroub, N., Ding, J., Zhang, W., Khan, F. F., Elhoseiny, M. *Wider Coverage, Better Plan: Admissible Alternatives for End-to-End VLA Planning* [Under review, ICLR 2027]
- **Felemban, A.**, Hroub, N., Ding, J., Bakr, E.M., Shen, X., Mohamed, A., Elhoseiny, M. *iMotion-LLM: Motion Prediction Instruction Tuning* [WACV 2026]
- Gou, C., **Felemban, A.**, Khan, F.F., Zhu, D., Cai, J., Rezatofighi, H., Elhoseiny, M. *How Well Can Vision Language Models See Image Details?* [ArXiv:2408.03940, 2024]
- **Felemban, A.\***, Battach, Y.\* (Equal contribution), Khan, F. F., Radwan, Y. A., Li, X., Marchese, F., Beery, S., Jones, B. H., Benzoni, F., Elhoseiny, M. *ReefNet: A Large-Scale Dataset and Benchmark for Fine-Grained Coral Reef Recognition* [NeurIPS 2026, Evaluations & Datasets Track, arXiv:2510.16822]
- Khan, F. F., Radwan, Y., Bakr, E.M., **Felemban, A.**, Mir, A., Michiels, N. K., Temple, A. J., Berumen, M. L., Elhoseiny, M. *FishNet 2.0: A Comprehensive Multimodal Benchmark for Long-Tailed Fine-Grained Fish Understanding* [Under review, ICLR 2027; earlier version arXiv:2509.25564]
- Al-Naffouri, T.Y., Masood, M., **Felemban, A.**, Aljoud, M.F., Bader, A., Alkhodary, M.T., Ahmed, T.B.K. *Real-Time Control of Mechanical Valves Using Computer Vision and Machine Learning* [US Patent App. 18/021,241, 2023]
- **Felemban, A.** *Deep Learning Action Anticipation for Real-Time Control of Water Valves: Wudu Use Case* [MS Thesis, 2021]
- AlSharif, M.H., Ahmed, M., **Felemban, A.**, Zayat, A., Muqaibel, A., Masood, M., Al-Naffouri, T.Y. *Robust 2D Indoor Positioning Algorithm in the Presence of Non-Line-of-Sight Signals* [EUSIPCO 2020]

## SELECTED PROJECTS

### DriveGavel: Zero-Shot VLM Safety Layer for Driving Planners

- Training-free VLM judge wrapping any planner: reads deterministic collision evidence, approves or overrides with a pre-verified evasive maneuver; collisions 83.3% → 13.3% on NeuroNCAP closed-loop.

### Wider Coverage, Better Plan (formerly iMotion-VLM): VLA Planning with Admissible Alternatives

- Trained a VLA planner's head on admissible alternative plans accepted by a rule-based scorer, plus an RL stage with speed-matched controls: NAVSIM v2 navhard single-plan EPDMS 0.315 → 0.373, best-of-16 0.359 → 0.614.

### iMotion-LLM, InstructNuPlan / InstructWaymo Datasets

- Built safety- and feasibility-aligned instruction datasets over vectorized BEV representations; trained and evaluated instruction adherence by coupling LLMs with trajectory-prediction modules.

### ReefNet (925K Annotations): Large-Scale Dataset and Benchmark

- 925K-annotation coral benchmark built with marine ecologists; ViT/BEiT/BioCLIP/CLIP/SigLIP baselines and a verification pipeline.

### Smart-Tap: Fully Integrated AI System for Water Valve Control

- Real-time action-anticipation prototype with pose estimation and spatio-temporal GCNs.

## EDUCATION

|            |                                                                                                                                                                                                                                                                                                                                                                                 |                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>PhD</b> | <b>King Abdullah University of Science and Technology (KAUST)</b> , Electrical & Computer Engineering, Saudi Arabia                                                                                                                                                                                                                                                             | 01/2022 - present |
|            | <ul style="list-style-type: none"><li>• <b>GPA:</b> 3.79/4.0</li><li>• <b>06/2023 - present:</b> Prof. Mohamed Elhoseiny. Vision-language-action planning and closed-loop safety evaluation for autonomous driving, and large-scale benchmarks (ReefNet)</li><li>• <b>01/2022 - 05/2023:</b> Prof. Tareq Al-Naffouri. Human action anticipation for real-time control</li></ul> |                   |
| <b>MS</b>  | <b>King Abdullah University of Science and Technology (KAUST)</b> , Electrical & Computer Engineering, Saudi Arabia                                                                                                                                                                                                                                                             | 01/2020 - 01/2022 |
|            | <ul style="list-style-type: none"><li>• <b>GPA:</b> 3.83/4.0</li><li>• <b>Advisor:</b> Prof. Tareq Al-Naffouri</li><li>• <b>Focus:</b> Deep Learning and Human Action Prediction</li></ul>                                                                                                                                                                                      |                   |
| <b>BS</b>  | <b>King Fahd University of Petroleum and Minerals (KFUPM)</b> , Electrical Engineering, Dhahran, Saudi Arabia                                                                                                                                                                                                                                                                   | 08/2014 - 12/2019 |
|            | <ul style="list-style-type: none"><li>• <b>GPA:</b> 3.43/4.0</li><li>• <b>Concentration:</b> Signal Processing and Wireless Communication</li><li>• <b>Senior Project:</b> Real-Time AI Surveillance System</li></ul>                                                                                                                                                           |                   |
| -          | <b>Georgia Institute of Technology</b> , Exchange Student, Atlanta, GA, USA                                                                                                                                                                                                                                                                                                     | 08/2018 - 12/2018 |

## SKILLS

**Multimodal / VLM / VLA:** Qwen-VL, LLaVA, CLIP/SigLIP, instruction tuning (SFT/DPO/GRPO), multimodal reasoning.  
**Autonomous Driving / Motion:** BEV and trajectory representations, perception-prediction-planning pipelines, open/closed-loop evaluation, Waymo/nuScenes/nuPlan workflows.  
**Training & Systems:** PyTorch, multi-GPU training, Slurm, vLLM inference, large-scale preprocessing.  
**Programming:** Python, OpenCV, MATLAB, R, TensorFlow (basic), C++ (basic).  
**Deep Learning:** Vision transformers, temporal and generative models, end-to-end AD systems, graph deep learning.  
**Additional Skills:** Data curation, metadata cleaning, parallel processing, digital signal/image processing.

## OTHER ACADEMIC CONTRIBUTIONS

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| <b>ELLIS Summer School on Autonomous Driving (SSAD 2026)</b> , Barcelona. Selected participant. Poster on DriveGavel and admissible-alternative VLA training. Discussions with Maximilian Igl, José M. Álvarez (NVIDIA), Holger Caesar (TU Delft), Kashyap Chitta, Alexandre Boulch (Valeo), Puneet Dokania (Five AI). | Sep. 15–18, 2026 |
| <b>Selected Student Presenter:</b> KAUST Rising Stars in AI Symposium 2026. Organizers: Jürgen Schmidhuber and Bernard Ghanem.                                                                                                                                                                                         | Feb. 9–11, 2026  |
| <b>Reviewer:</b> WACV 2026, ICCV2025 Workshop Proceedings, ICCV 2025 Workshop CV4E, ACML 2024, SIGGRAPH Asia 2024.                                                                                                                                                                                                     |                  |
| <b>Workshop Organizer:</b> CVPR 2024 – C3DV: 2nd Workshop on Compositional 3D Vision.                                                                                                                                                                                                                                  | 06/2024          |

## AWARDS & FUNDING

**CEMSE Dean's List Award 2021/22** (top 20% of students)  
**\$1M in Research Funding:** NEOM Ocean Science and Solutions (OSSARI) Research Fund (\$600k, 2024); KAUST Translational Fund (\$450k, 2022).  
**Projects Awards:** Start Smart [🔗](#) (1st place, \$9k, EcoStream, 2024), Digital Innovation Award [🔗](#) by the Saudi Ministry of Communications and IT (1st place, Smart-Tap, \$21k, 2022), KFUPM Undergraduate Senior-Project Award (1st place, AI Surveillance Project, 2019), SSI Internship Poster Competition (2nd place, Indoor localization project, 2019).

## EXPERIENCE

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|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>KAUST Academy 1-Week AI Specialization Instructor:</b> Taught undergraduates the fundamentals of AI (ML, CV, neural networks, Python). | 01/2025           |
| <b>KAUST Saudi Summer Internship (SSI):</b> Ultrasonic indoor localization, filtering non-line-of-sight signals with machine learning.    | 05/2019 - 08/2019 |
| <b>KFUPM IEEE Student Branch Publicity Chair:</b> events and seminars.                                                                    | 01/2018 - 01/2019 |